MPhil Thesis: Data

Mohammad-Shah Karim

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Contents

1 Data

1.1 Overview and Unit of Observation

The empirical analysis draws on five sources: the Medicaid State Drug Utilisation Data (SDUD) maintained by the Centers for Medicare and Medicaid Services (CMS); the Drugs@FDA database; patent bulk data files distributed by the USPTO through the PatentsView project; the Texas Vendor Drug Program formulary; and the National Library of Medicine's RxNorm and RxClass systems, which provide the therapeutic classification bridge across sources. Table 1 summarises the contribution of each source to the final dataset.

Table 1: Data sources and their roles in the analysis

Source	Key identifier	Role
SDUD (CMS)	NDC, state, year, quarter	Medicaid and non-Medicaid prescription volumes and reimbursement
Drugs@FDA	Application number, sponsor name	FDA drug approval outcomes
USPTO / PatentsView	Patent ID, assignee name	Patent counts by therapeutic class
Texas Vendor Drug Program	NDC, effective date	PDL status and prior authorisation indicators
RxNorm / RxClass	RxCUI, ATC code	Therapeutic classification of NDC and active ingredients

The final estimation dataset is a balanced panel at the firm × ATC Level 1 × year × quarter level, spanning 2010 to 2025. The sample period begins in 2010 because the pre-expansion Medicaid exposure measures require at least two years of pre-reform prescription data, and the SDUD is downloaded from 2005 onwards for this purpose. All datasets are harmonised to the World Health Organisation's Anatomical Therapeutic Chemical (ATC) classification system at Level 1, as described in Section **??**. The construction of treatment and outcome variables, including the Medicaid exposure measures and PDL exposure variables, is described in the Methodology section.

1.2 Medicaid State Drug Utilisation Data

The main source for Medicaid prescription demand is the State Drug Utilisation Data (SDUD), maintained by the Centers for Medicare and Medicaid Services (CMS). The SDUD was established alongside the Medicaid Drug Rebate Program and records outpatient prescription drugs reimbursed by state Medicaid programmes at the drug × state × quarter level. It covers all fifty states and the District of Columbia. Each record is identified by an 11-digit National Drug Code (NDC) and reports prescription counts, units reimbursed, and reimbursement amounts split into Medicaid and non-Medicaid components. Annual files covering 2005 to 2024 are downloaded via the Medicaid Data portal API, yielding a raw dataset of 42,031,131 records.

The SDUD plays two roles in the analysis. First, it is used to construct Medicaid exposure measures at the therapeutic-class - this forms the main treatment variable for the analysis. Secondly, once merged with PDL information from the Texas Vendor Drug Program, the SDUD is used to construct exposure measures to formulary placement and pricing pressure at the firm-level.

Table 2: Key variables in the SDUD

Variable	Description
NDC	11-digit code identifying each drug product, split into a 5-digit labeler code, 4-digit product code, and 2-digit package code
Drug name	Proprietary or generic name of the drug
Units reimbursed	Number of units reimbursed by Medicaid in the state-year-quarter cell
Number of prescriptions	Number of prescriptions filled for a given drug in the state-year-quarter cell
Reimbursement amount	Total reimbursement for a given drug, split into Medicaid and non-Medicaid components

1.2.1 National Drug Code structure

The NDC is an 11-digit product identifier - this contains a 5-digit labeler code, a 4-digit product code, and a 2-digit package code. The SDUD stores the NDC as an 11-digit string without hyphens. However, throughout this analysis the labeler, product, and package components are extracted by splitting on character position (positions 1-5, 6-9, 10-11). External sources that use the hyphenated three-segment format such as the FDA NDC directory require the inverse transformation where each raw segment is zero-padded to its standard width before concatenation. This normalisation is applied uniformly across all datasets to ensure that NDC-based joins do not introduce incorrect matches due to formatting differences.

1.2.2 Firm identification

The SDUD does not record firm names, so firms are instead identified through their NDC labeler code. However, pharmaceutical firms commonly operate with the use of multiple labeler codes due to subsidiaries or acquisition arrangements. As a result, labeler codes alone cannot be treated as a unique firm identifier in this analysis.

A firm mapping is constructed from the FDA NDC directory. Each labeler code is initially linked to its registered labeler name from the FDA's product and package text files. These names are then clustered into firm-level groups using a text similarity procedure. Labeler names are standardised by converting to lowercase, removing punctuation, and stripping trailing whitespace. Unigram tokenisation is then used to construct a term frequency-inverse document frequency (TF-IDF) representation, and pairwise cosine similarity is computed across all cleaned labeler names. Pairs with a cosine similarity score above 0.85 are treated as matches and are grouped into firm clusters, so that labeler codes with sufficiently similar names are assigned to the same firm. Each cluster is assigned the name of its first member as the canonical firm identifier. Merging the resulting firm mapping into the SDUD by labeler code allow for 40,003,656 of 42,031,131 records to match, which corresponds to a match rate of 95.18%.

1.2.3 Therapeutic classification

Each NDC in the SDUD is mapped to an ATC Level 1 class using the RxNorm API. First, each NDC is submitted to the RxNorm NDC status endpoint in order to obtain its RxCUI identifier. This is then used to retrieve ATC codes through the RxNorm ATC lookup endpoint. ATC Level 1 is defined by the first character of the ATC code, while ATC Level 2 is defined by the first three characters. This procedure classifies 96.10% of SDUD records. Records that cannot be classified are excluded from the estimation sample. In cases where a single NDC maps to more than one ATC class, the record is duplicated so that each NDC-class combination appears separately. These duplicated records are then aggregated within class cells in the final panel.

1.3 Texas Preferred Drug List Data

PDL data is taken from the Texas Vendor Drug Program, which publishes machine-readable historical records of Medicaid drug coverage decisions. Texas is selected because it is one of very few states to make its complete formulary history available in a machine-readable format with drug-level effective and end dates. Unlike many other states, whose PDL data are available only as static PDFs or current cross-sections, the Texas data allow a time-varying measure of drug-level preferred status to be constructed. The formulary file records the following fields for each listed drug: the NDC, the Medicaid coverage start and end dates, a coverage status indicator, a flag indicating whether PDL-related prior authorisation is required, and a flag indicating whether clinical prior authorisation is required.

Two variables are constructed to separate the clinical and pricing margins of PDL selection. PT_Stage_Pass captures drugs that clear the Pharmacy and Therapeutics clinical review, i.e. these are drugs which are clinically acceptable. Price_Stage_Pass captures drugs that clear the pricing stage, meaning that their net cost after rebates is sufficiently competitive for preferred placement. A drug is therefore classified as being on the PDL when it is covered by Medicaid and clears both the clinical and pricing stages.

1.3.1 Effective date treatment

Coverage start and end dates are recorded at the month-day-year level. A drug is classed as having active coverage in a given year-quarter if its coverage start date falls prior to the end of the quarter and either no coverage end date is recorded or the end date falls after the start of the quarter. Drugs for which no end date is present, or for which the end date is in 2025 or later, are treated as currently active.

1.3.2 Linking to the SDUD

The Texas formulary is merged to the SDUD at the full 11-digit NDC level - it is matched on the labeler, product, and package codes simultaneously. NDC records without a corresponding Texas formulary entry receive missing values for their PDL indicators as opposed to being excluded. Missing PDL values are subsequently set to zero when constructing the firm-level PDL exposure measures - absence from the Texas Drug Vendor

Program record likely reflects a drug that has not obtained preferred status rather than a data gap.

1.3.3 Representativeness

Since the PDL analysis relies on a single state's formulary, the extent to which the Texas PDL is representative of the Medicaid PDL choices in other states is assessed by computing pairwise overlap rates with comparison states. This representativeness analysis is limited by data availability - the majority of states do not publish machine-readable PDL records with historical coverage dates, so the comparison can only be conducted for the subset of states whose formulary data are available in a structured format. On this basis, New York, California, and Washington are used as comparison states. Overlap is measured in both directions as the share of drugs on one state's PDL that also appear on the other state's PDL, matched at the labeler and product level of the NDC.

Table 3: Pairwise overlap of state Preferred Drug Lists

State 1	State 2	Overlap (State 1 in State 2)	Overlap (State 2 in State 1)
New York	Texas	0.375	0.950
California	Texas	0.196	0.197
Washington	Texas	0.223	0.812
California	New York	0.722	0.230
California	Washington	0.384	0.106

The results indicate that 95.0% of drugs on the Texas PDL also appear on the New York PDL, and 81.2% also appear on the Washington PDL. Overlap with California is substantially lower at around 20% in both directions, reflecting California's distinct formulary structure. The overlap analysis suggests that the Texas PDL shares is somewhat representative of the preferred drug lists of other large states. However, it should not be treated as a direct proxy for every state's formulary decisions, especially given the weaker overlap with California. In this thesis, the Texas PDL is used primarily to capture the two-stage clinical-and-price structure of PDL selection, rather than to measure the exact set of preferred drugs nationwide.

1.4 FDA Approval Data

Approval data for drugs is obtained from the Drugs@FDA database, which records all drug applications submitted to the FDA. Three files are specifically used for the purpose of this study. Applications are used to determine the sponsor name and application number of a given drug, the submissions file provides the FDA's action taken on a given outcome, and the products file records the active ingredients and drug names associated with each application number.

1.4.1 Application types

The Drugs@FDA dataset contains three types of applications: New Drug Applications (NDAs), Abbreviated New Drug Applications (ANDAs), and Biologics License Applications (BLAs). While NDAs and ANDAs count towards approvals of novel drugs, ANDAs correspond to generic drug approvals. This paper proposes to retain ANDAs within the set of innovation applications as generic entry of drugs is a form of product approval that can affect the competitive environment within a class. The sample is restricted to approvals occurring on or after 1 January 2005 to align with the SDUD coverage window.

1.4.2 Therapeutic classification

Therapeutic classification of FDA-approved drugs occurs at the ingredient level. To do this, active ingredients strings are standardised and split to account for multi-ingredient products. Each combination of normalised ingredients are then queried against the NLM's RxClass API. This maps each ingredient to one or more ATC codes. Ingredient-level results are collapsed at the application level so that each application is linked to the union of ATC Level 1 classes associated with its active ingredients. Applications for which no ATC class can be assigned are excluded. This procedure classifies 48,143 of 50,368 application-product observations giving a match rate of 95.58%.

In cases where an application maps to multiple ATC Level 1 classes, the application is counted once within each applicable class at the aggregation stage. This is analogous to the treatment of multi-class NDCs in the SDUD.

1.4.3 Firm-NDC matching

Since the Drugs@FDA database identifies firms by their name as opposed to their NDC labeler code, an additional matching step is required. FDA sponsor names and NDC labeler names are standardised by: conversion to uppercase, removal of non-alphanumeric characters, and stripping of common corporate suffixes such as Inc, Corp, LLC. A three-stage merging procedure is then applied. In the first stage, firms are matched exactly by their cleaned name. In the second stage, unmatched FDA sponsors are linked to NDC labeler names via a root-token match, where the root is the first alphanumeric token which is at least four characters long. Finally, remaining unmatched firms are linked using Jaro-Winkler string similarity at a threshold of 0.90, with the highest-scoring NDC match being used. Under this process, a match rate of 82.26% is achieved.

1.5 Patent Data

Patent data is obtained from the USPTO PatentsView data files. PatentsView harmonises patent records and provides disambiguated assignee information. Specifically, for this study, five files are used: the base patent file, which provides grant dates; the current CPC classification file; the disambiguated assignee file; the CPC group title file; and the WIPO technology field classification file.

1.5.1 Filtering for relevant patents

The universe of patents is filtered in two steps. First, patents are restricted to those classified under the WIPO technology field "Pharmaceuticals". This provides a broad filter for patents associated with pharmaceutical products, including biological and chemical technologies. Within this set, patents are restricted to those that fall under the "A61P" subclass - this allows for the study to focus on patents with a stated therapeutic use. Patents classified in A61P but not in the WIPO Pharmaceuticals field are also included, as some patent-level WIPO assignments may be incomplete. The A61P covers the therapeutic activity of chemical compounds or medicinal preparations, therefore this is the most relevant CPC subclass for mapping patents to therapeutic classes. Patents classified under A61P are kept even if they are not assigned to the WIPO pharmaceutical field, since some patent-level WIPO assignments may be incomplete. This allows for the study to focus on patents that are most likely to be related to therapeutic drug innovation.

It is important to note that the CPC classifications are not perfect measures of pharmaceutical invention of a patent, since CPC classes are assigned during the patent examination process. Consequently, the A61P restriction may exclude some relevant patents relating to delivery or formulation, while also retaining some patents that are classified as pharmaceutical but less directly related to new therapeutic agents.

1.5.2 Therapeutic classification

The A61P subclass is divided into CPC groups corresponding to different therapeutic uses. A manually constructed crosswalk is used map each A61P group to an ATC Level 1 class, with the crosswalk following the structure of the USPTO's A61P classification. Selected entries from this crosswalk are shown in Table 4. For example, patents related to cardiovascular activity are classified under A61P9, so these are mapped to ATC class C, while patents related to nervous system activity fall under A61P25, thus they are mapped to ATC class N. In cases where a patent falls under multiple different A61P CPC groups, it is counted once within each applicable class.

Table 4: CPC A61P to ATC Level 1 crosswalk (selected entries)

CPC Group	Therapeutic area	ATC Level 1
A61P1	Gastrointestinal	A
A61P7	Blood	B
A61P9	Cardiovascular	C
A61P17	Dermatology	D
A61P15	Genito-urinary and sex hormones	G
A61P5	Endocrine	H
A61P31	Anti-infectives	J
A61P35	Antineoplastics	L
A61P19	Musculo-skeletal	M
A61P25	Nervous system	N
A61P33	Antiparasitic	P
A61P11	Respiratory	R
A61P27	Sensory organs	S

1.5.3 Firm–NDC matching

Patent assignee names are matched to the NDC labeler firm clusters using the same three-stage procedure applied to sponsors in the Drugs@FDA dataset. Cleaned assignee names are first matched to labeler names exactly. The remaining unmatched assignees undergo root-token matching followed by a Jaro-Winkler fuzzy match at a threshold of 0.90.

Out of 28,143 unique assignees, 7,566 were matched to the NDC firm universe, which corresponds to a match rate of 27.1%. The patent assignee universe includes academic institutions, research hospitals, individual inventors, and non-pharmaceutical entities that may hold A61P patents but have no commercial presence in the Medicaid drug market. These entities do not have labeler codes within the SDUD. Additionally, pharmaceutical firms may assign patents to research subsidiaries whose names differ from the trading names under which drugs are marketed. As a result, the match rate is substantially lower than the Drugs@FDA match rate.

Unmatched patent assignees are kept in the patent count data, but they cannot be linked to firm-level PDL exposure. These sponsors are therefore excluded from specifications that require firm-level linkage to the SDUD.